

[Language Translators]/ [Third]

In this lab we start to create our project by using oop in java
[First Semester][2025-2026]



[Lab 5] 1- Create new package (compiler)

[08:30 Am]

2- The first class in this package (main class) → class compiler:- in this class we call the other classes , methods and print the result.

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[2 Hours]

```
package compiler;

public class Compiler {

    public static void main(String[] args) {
        Look_up_table Lutt=new Look_up_table();
        Lexical L=new Lexical();
        L.run(Lutt);
        for(int i=0;i<Lutt.size;i++)
            System.out.println(Lutt.getLex(i)+"      Type is      "+Lutt.getTyp(i));
    }
}
```

3- The second class is (Look_up_Rec) that has

- Two var (lexem,type) as a private string
- Method (create_look_up_rec)
- Method (getLexem) that return the lexem from record
- Method (getType) that return the Type from record

```
package compiler;

public class Look_up_rec{
    private String lexeme;
    private String type;

    public void creat_Look_up_rec(String l,String t){
        lexeme=l;
        type=t;}

    public String getLexeme(){
        return(lexeme);
    }

    public String getType(){
        return(type);
    }
}
```

4- The third class is (Look_up_Table) that has

- Define an array of the previous class of record as public
- Define the index of array as public int
- Initialize the Look_up_Table class
- Method (insert) which is open new record in the lut class then call the (create_look_up_rec) and inc the index by one.
- Method (getLex) that return the lexem from Table
- Method (getTyp) that return the Type from Table

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package compiler;

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public class Look_up_table {

[Lab 5] public static Look_up_rec[] Lut=new Look_up_rec[1001] [08:30 Am]

public int size;

Look_up_table(){size=0;}

Instructor Information

Instructor public void insert(String l,String t)

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[Ibtihal Abbas] Lut[size]=new Look_up_rec();

Lut[size].creat_Look_up_rec(l, t);

size++;

}

public String getLex(int i)

{

return(Lut[i].getLexeme());

}

public String getTyp(int i)

{

return(Lut[i].getType());

}

}

5- The fourth class is (Lexical) that has

- Define an object of the Look_up_Table class as public
- Method Insert_lut → (RW,Dig,ID)
- Method Insert_ch →(Sp,Op)
- Method Run that has an object of Look_up_Table

Notice // the run method it's the old main method but we must remove the printing of the table from it.

package compiler;

import java.io.*;

import java.util.*;

public class Lexical {

public Look_up_table Lut=new Look_up_table();

public static boolean is_SP(char ch){

if((ch==' ')|| (ch==';') || (ch=='(') || (ch==')') || (ch=='{') || (ch=='}'))

return(true);

else return(false);

}

public static boolean is_OP(char ch){

if((ch=='+')|| (ch=='-')|| (ch=='*')|| (ch=='/')|| (ch=='='))

return(true);

else return(false);

}

public static boolean is_RW(String tok){

if(("DEF".equals(tok)) || ("BODY".equals(tok)) || ("int".equals(tok)) ||

("byte".equals(tok)) || ("print".equals(tok)))

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```
return(true);
else return(false); }
```

```
public static boolean is_DIG(String tok){
    int i;
```

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Hours

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```
return (true);
}

public static boolean is_ID(String tok){
    int i;
    if((tok.charAt(0)>='a'&& tok.charAt(0)<='z') ||
        (tok.charAt(0)>='A'&& tok.charAt(0)<='Z'))
    {
        for (i=1;i<=tok.length()-1;i++){
            if ((tok.charAt(i)>='a'&& tok.charAt(i)<='z') ||
                (tok.charAt(i)>='A'&& tok.charAt(i)<='Z') ||
                (tok.charAt(i)>='0'&& tok.charAt(i)<='9'))
                continue;
            else
                return (false);
        }
    }
    else
        return (false);
    return (true); }
```

```
public void insert_Lut(String tok){
    if (is_RW(tok))Lut.insert(tok,"RW");
    else if(is_DIG(tok))Lut.insert(tok,"DIG");
    else if(is_ID(tok))Lut.insert(tok,"ID");
    else System.out.println(tok+" it is Error");
}
```

```
public void insert_ch(char c){
    if (is_OP(c)){Lut.insert(String.valueOf(c),"OP");}
    else if(is_SP(c)){Lut.insert(String.valueOf(c),"SP");}
    else System.out.println(c+" it is Error");
}
```

```
public void run(Look_up_table Lut1)
{
    Scanner in;
    Lut=Lut1;
    try {
        in = new Scanner(new File("C:\\Users\\khyro\\Desktop\\nt.txt"));
        char ch;
        String line, tk;
        while(in.hasNext()){
            line = in.nextLine();/*read one line from file*/
            tk = "";
            for(int i=0; i<line.length(); i++){
                ch = line.charAt(i);
                if (ch!=' ')
                {
                    if(is_SP(ch)||is_OP(ch)){/* must be send to LUT*/
                        if (!"".equals(tk)){
```

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[First Semester] [2025-2026]

[Lab 5]

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```

        insert_Lut(tk);
        tk="";
        insert_ch(ch);}
    else
        tk+=ch;
}
else if (!"".equals(tk)){
    insert_Lut(tk);
    tk="";
}
}/*end for loop*/
if (!"".equals(tk)) {insert_Lut(tk);tk="";}
}/*end while loop*/
in.close(); /*close text file*/
}/*end main1 method*/
catch(Exception e){}
    }
```